From Words to Desired Images: Language-Specified Goal Views for Visual Servoing in Gaussian Splatting Scenes

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Abstract

Visual servoing needs a desired image, which is normally captured or chosen by an operator. We generate it from a text query in a scene reconstructed with 3D Gaussian Splatting (3DGS). The queried object is grounded with an open-vocabulary detector and a promptable segmenter, lifted to 3D with the rendered depth, and summarised by its centroid, extent and dominant plane. A geometric viewpoint selection then renders candidate views and scores their visibility, render reliability (training-camera coverage), a frontality or broadside prior and whether the whole object fits in the image with the camera in free space; the render at the chosen pose is the desired image. On the 67 labelled queries of LERF-OVS, scored against ground-truth 3D objects, our desired images are good views (object visible, whole, centred, well sized) in 30% of the queries, against 9% for CLIP render-and-score and 12% for CapFrame, a recent text-to-camera method based on multimodal large language models, and they are always level. The training photograph in which the object was detected reaches 21%, a difference that is not significant, but it centres the object less often (30% against 48% of the queries). A photometric visual servo started from perturbed poses reaches our desired images in 85% of the runs (76% for larger perturbations), more often than those of CLIP render-and-score and CapFrame, and than the training photograph from small perturbations. Grounding is the bottleneck: when the detector finds the right object, 62% of our views are good; with ground-truth masks, 69% of all queries get a good view. Videos and results: https://text2desired.github.io/.

Keywords: Visual servoing, Gaussian splatting, open-vocabulary perception, viewpoint selection, Desired image generation

1 Introduction

Image-based visual servoing (IBVS) drives a camera so that visual features reach desired values, classically extracted from a desired image [1]. The task is therefore specified by an image that someone must capture or select beforehand. When the target is described by language ("look at the teapot"), no such image exists. Radiance-field scene representations change this situation: a 3D Gaussian Splatting scene [2] renders a photorealistic image from any pose, so the desired image can be *generated* if a good goal pose is known.

Choosing that pose is the difficult part. Language-embedded radiance fields [3–6] localize objects from text but do not propose a viewpoint. Visual servoing with radiance fields [7] or foundation-model features [8, 9] still assumes the desired image is given, and language-driven servoing either only selects the target object or hallucinates a 2D goal with a diffusion model [10]. The closest work, CapFrame [11], grounds free-form photographic instructions to a camera pose in a 3DGS scene with multimodal large models. It asks whether the image matches the words; a servo target must also be reachable within the robot's degrees of freedom (level, in free space) and framed so that a servo converges to it, which CapFrame does not consider.

We address object-centric goal views for visual servoing and evaluate them both as images and as servo targets. Our contributions are:

- a training-free pipeline that turns a text query into a desired image in a 3DGS scene: zero-shot grounding, depth back-projection and a geometric viewpoint selection that fits the whole object in free space, prefers well-reconstructed views and keeps the horizon level;
- a task-level benchmark on LERF-OVS [4] that scores desired images against ground-truth 3D objects (visible, whole, centred, well sized, level) and by the success of a photometric servo [12] toward them;
- a comparison with retrieval, CLIP render-and-score and CapFrame [11], an analysis that separates grounding errors from viewpoint errors (detector, language field and ground-truth masks), and a score ablation.

The supplementary material reports the servo across reconstructions and on Mip-NeRF 360, the sensitivity to the parameters and a ground-robot variant in synthetic houses; videos are on the project page, https://text2desired.github.io/.

2 Related Work

Visual servoing and goal specification. IBVS and PBVS [1] regulate image features or poses toward a desired configuration; image moments [13, 14] and photometric features [12] avoid explicit correspondences but still require a desired image. Deep and foundation-feature servoing [8, 9] likewise align to a reference image, and NeRF-based servoing [7] renders the current view while the goal is given. Pose-based alternatives with novel-object pose estimators [15, 16] need object models and a manually defined goal pose. Imagine2Servo [10] generates goal images with a 2D diffusion model, without 3D consistency. Object-centred servo tasks that keep the object centroid at the image centre and at a fixed distance are classical; Misimi *et al.* [17] use such a task as the

primary task of an active-vision scheme and explore the object in its null space for 3D reconstruction. Our final-approach features (centroid, distance, plus viewing direction and roll) are related, but our goal is a single desired view chosen before servoing.

Language in radiance fields. LERF [3], LangSplat [4], LangSplatV2 [5] and Dr. Splat [6] embed CLIP [18] features to localize objects from text, as relevancy maps or per-Gaussian scores. They output where the object is, not how to look at it. ChatCam [19] generates camera trajectories from conversation, and Retri3D [20] moves cameras toward artifact-free regions to retrieve scenes.

Text-instructed viewpoints: CapFrame. CapFrame [11] is the closest work: it places a camera in a 3DGS scene from a free-form photographic instruction. A multimodal large language model (MLLM) ranks the training views, translates the instruction into layout boxes and, for objects it judges to have a front, an orientation target estimated with Orient Anything [21]; the pose is then refined by differentiable rendering. Our problem differs in what the image is for: a photograph must match the words, a desired image must be a target that the robot can reach and that a servo converges to. Our servo reaches CapFrame's images less often than ours: on the 61 queries where it returns a view, 66% against 85% of the runs from small offsets and 60% against 76% from large ones (Wilcoxon signed-rank test on the per-image success, $p < 0.001$ and $p = 0.004$). Its object is centred in 21 of its 61 views, and 84% of them are rolled by more than 5° (median 21°), which a robot that cannot roll its camera cannot reproduce. Our selection keeps the horizon level, centres the object with the whole of it in view and keeps the camera in free space. It obtains the object's orientation from PCA of the lifted cloud instead of an MLLM and chooses by an explicit score over rendered candidates, so it is training-free, deterministic and runs on a 24 GB GPU in 35–75 s per query (CapFrame: 136–518 s, median 312 s, on the same GPU with 4-bit weights). CapFrame is the more expressive method: it follows compositional instructions and its MLLM knows names that our detector does not. The two are complementary: an MLLM could supply a semantic front or a compositional constraint as a term of S, while the other terms keep the view usable as a robot goal.

Viewpoint quality. Human studies of canonical views [22, 23] show strong agreement on preferred views, often three-quarter rather than frontal for volumetric objects. Object-goal navigation counts success by reaching the object [24] and does not score the final view. Render reliability in 3DGS can be estimated with uncertainty methods such as FisherRF [25]; we use training-camera coverage as a cheap proxy.

3 Method

3.1 Grounding and geometry

Given a query q, Grounding DINO [26] scores every training view; the best-scoring view whose detection box does not touch the image border is preferred, but only if it shows the same object as the overall best detection (the 3D point of the best box center, back-projected with rendered depth, must project inside its box). This avoids truncated objects without replacing a true detection by a false one. SAM [27] segments the box; the eroded mask is back-projected with the 3DGS expected depth, depths outside the 5–95% range are discarded, the cloud is voxel-downsampled to a uniform

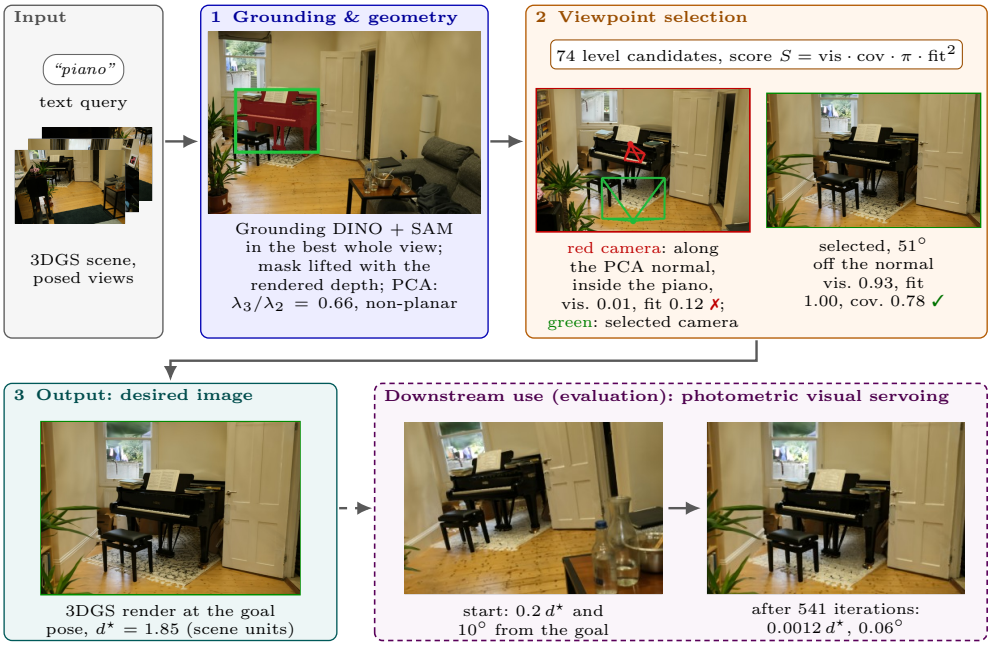


Fig. 1 Overview on the query "piano" (Mip-NeRF 360 *room*). (1) The query is grounded in the best detection view that shows the whole object (red: SAM mask, green: detection box); the mask is lifted to 3D with the rendered depth and PCA gives the centroid, the dominant-plane normal and the planarity ratio. (2) The piano is not planar and its PCA normal points into it: a camera along the normal would be inside the object (red camera, seen from outside; green: the selected camera). Level candidates are rendered and scored by S (visibility, coverage by the training views, shape prior π and fit; Fig. 2), and the selected one fits the whole piano in free space and is well covered by the training views (green). (3) Its 3DGS render is the desired image, the output of the method, with the goal pose. Dashed: a downstream use, as in our evaluation; a photometric visual servo, started 0.2 $d^\star$ and 10° away, reaches it.

density and cleaned by statistical outlier removal. PCA of the cloud gives the centroid $\boldsymbol{p}_o$, the eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ and the normal $\boldsymbol{n}_o$ (smallest eigenvector), oriented toward the reference camera. The planarity ratio $\rho = \lambda_3/\lambda_2$ measures how well a dominant plane exists. Any other source of a 2D mask in a training view can replace this step; we evaluate a language field and ground-truth masks in Sec. 4.3.

3.2 Viewpoint selection and desired image

Candidate viewing directions $\boldsymbol{v}$ (from the object to the camera) are the PCA normal, the reference-view direction and 72 level directions (24 azimuths at the 25/50/75% percentiles of the training-camera elevations seen from the object); the PCA normal is dropped for non-planar objects, and directions whose required distance is undefined are skipped. For each $\boldsymbol{v}$, the camera looks at $\boldsymbol{p}_o$ with a level horizon. The required distance $d_{\text{req}}(\boldsymbol{v})$ makes the object's horizontal and vertical extents (seen along $\boldsymbol{v}$) span at most a fraction $\phi = 0.6$ of the image; $d_{\min}(\boldsymbol{v})$ is the shortest distance at which they still fit in the image (fraction 0.85); the free distance $F(\boldsymbol{v})$ is the rendered depth from

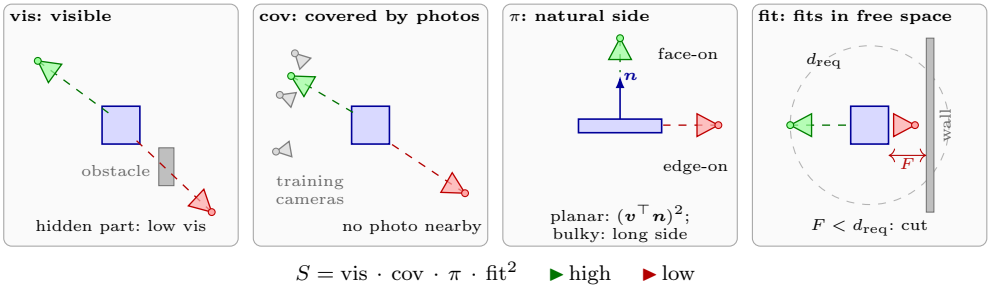


Fig. 2 The four terms of the viewpoint score, seen from above: for each, a candidate camera with a high value (green) and one with a low value (red). vis: fraction of the object visible, not occluded; cov: closeness to a training camera, where the 3DGS render is reliable; π: a natural side, face-on for a planar object, the long side for a bulky one; fit: enough free space behind the camera to see the whole object.

the object outward along $\boldsymbol{v}$. The goal distance is

$$d(\boldsymbol{v}) = \min\big(d_{\text{req}}(\boldsymbol{v}),\, \mu F(\boldsymbol{v})\big), \qquad \text{fit}(\boldsymbol{v}) = \min\big(1,\, \mu F / d_{\min}\big), \tag{1}$$

with $\mu = 0.9$: the camera stays in free space, and fit drops below 1 when even the closest distance that shows the whole object is not free. Each candidate is rendered at distance $d(\boldsymbol{v})$ and scored by

$$S(\boldsymbol{v}) = \text{vis}(\boldsymbol{v})\, \text{cov}(\boldsymbol{v})\, \pi(\boldsymbol{v})\, \text{fit}(\boldsymbol{v})^2, \tag{2}$$

where vis is the fraction of object points visible in the render (not occluded in the rendered depth), cov is the coverage by the best-matching training camera (the largest product of Gaussian kernels on angle and log-distance over the training cameras that see the object), and $\pi = (\boldsymbol{v}^\top \boldsymbol{n}_o)_+^2$ for planar objects ($\rho < 0.2$) or $\frac{1}{2} + \frac{1}{2} b(\boldsymbol{v})$ otherwise, with b the alignment with the object's horizontal broadside. For a planar object, the normal is kept if its view is visible (≥ 0.6), covered (≥ 0.25) and fits (≥ 0.98); otherwise the best-scoring candidate among those that fit (≥ 0.98, or all if none does) is chosen, ties (all scores zero) being broken by $\text{vis}\,\text{cov}\,\text{fit}^2$. The render at the selected pose is the desired image $I^\star$, and the goal pose and distance $d^\star$ are kept with it.

3.3 Downstream use: reaching the desired image

The method ends with the desired image $I^\star$, together with the goal pose and the distance $d^\star$; it does not depend on how the robot reaches them. Any image-based servo can take $I^\star$ as its target, and a pose-based controller can take the goal pose. The servo is therefore not part of the method but the test of its output: a good desired image must be a target that a servo reaches. In the evaluation we use photometric visual servoing [12], because it sees only images (no features, no goal depth, no goal pose), so its success depends on the desired image alone: the error is $\boldsymbol{e} = I - I^\star$ on grey levels, the interaction matrix $\boldsymbol{L}_I = -\nabla I^\top \boldsymbol{L}_{\boldsymbol{x}}$ uses the current depth, and the Levenberg–Marquardt command $\boldsymbol{v}_c = -\lambda(\boldsymbol{H} + \mu\, \text{diag}\, \boldsymbol{H})^{-1} \boldsymbol{L}_I^\top \boldsymbol{e}$, $\boldsymbol{H} = \boldsymbol{L}_I^\top \boldsymbol{L}_I$, is integrated on

SE(3). The camera is free-flying: all six degrees of freedom are controlled, and the commanded velocity is applied exactly (no robot dynamics or joint limits).

4 Experiments

4.1 Setup

Scenes. LERF-OVS [4]: four scenes (figurines, teatime, ramen, waldo_kitchen), 67 text queries with polygon labels in a few frames each, images of about 988×730. Mip-NeRF 360 [28]: six scenes, 27 queries without labels, used to set the parameters and, in the supplementary material, for servoing and sensitivity. Each scene is a gsplat 3DGS model trained on all images for 30k iterations, without depth prior. All parameters were set on the Mip-NeRF 360 *room* scene and frozen for every other scene. One rule was added afterwards, the tie-break between candidates whose scores are all zero, after a failure on the *bicycle* scene; it changes none of the LERF-OVS results. The reconstructions are in the normalised units of each scene, not in metres; all distance tolerances are therefore relative to the goal distance $d^\star$. Detector Grounding DINO-tiny (threshold 0.2), SAM ViT-H. On an RTX A5000 (8 LERF-OVS queries), grounding takes 0.23 s per training view, dominated by the detector (30–68 s for 131–299 views), viewpoint selection 3.3–4.1 s, and one servo iteration 12 ms at 320 px; the desired image is computed once per query, before the servo starts.

Ground truth and criteria. For each query, every labelled polygon is eroded by 2 px, back-projected with the rendered depth (depths outside the 5–95% percentiles dropped), merged over the label frames and split into instances by DBSCAN (clusters below 5% of the points dropped; 87 instances for the 67 queries). A desired image is scored against the instance it shows best (largest visible fraction, the same rule for every method): *visible* (fraction of the instance's points in the image and not occluded, ≥ 0.5), *whole* (fraction inside the image, ≥ 0.9), *centred* (centroid at most 15% of the image diagonal from the image centre), *size* (larger projected extent, twice the 99th percentile of the points' offsets from the centroid, between 15% and 90% of the image) and *level* (horizon tilt below 5°). A *good view* passes the first four tests. Our grounding finds the *right object* when its 3D centroid lies in the 2–98% bounding box of a ground-truth instance or within 15% of the instance's extent from it. Methods are compared on the same queries, with the exact McNemar test for per-query outcomes and the Wilcoxon signed-rank test on per-image servo success (the runs of an image are not independent).

Baselines. *B1, detected training photo*: the training image in which our grounding detects the object (same detector, no view synthesis). *B2, CLIP render-and-score*: from every training camera, 5 yaws $\times$ 3 pitches with a level horizon are rendered and the view with the highest CLIP ViT-L/14 similarity to "a photo of a q" is kept, as the render-and-score baseline of [11]. *B3, CapFrame* [11]: the authors' code with Qwen3-VL-32B (4-bit NF4 weights to fit a 24 GB GPU) and the instruction "A view of the q, in the center of the frame."; it runs on the RGB Gaussians of LangSplat-V2 for the same scenes and its pose is converted to our frame. It fails on 6 queries (no candidate kept by its MLLM, empty mask, crash), counted as misses. Its authors evaluated it on the same four LERF-OVS scenes, with hand-written compositional instructions (3–4

Table 1 Desired images on LERF-OVS: percentage of queries passing each criterion (good view = visible, whole, centred and size; 95% Wilson interval in brackets) and photometric servo success over the runs (small / large perturbation), over all 67 queries and over the 29 queries whose object our grounding (Grounding DINO + SAM) finds, which separates our viewpoint selection from grounding errors (for ours and B1, which share the detection; for B2 and B3 it is only a subset of queries). Queries without a desired image (CapFrame's 6 failures) count as failures in every column, including their 6 servo runs.

method	good view	visible ≥50%	whole in frame	centred	size 0.15–0.9	level <5°	servo success small	servo success large
all 67 queries								
B1 detected training photo	21 [13,32]	**58**	**70**	30	**63**	67	70	71
B2 CLIP render-and-score	9 [4,18]	27	36	19	60	**100**	59	56
B3 CapFrame [11]	12 [6,22]	40	66	31	55	15	60	54
Ours	**30** [20,42]	40	60	**48**	61	**100**	**85**	**76**
right object found (29 queries)								
B1 detected training photo	41 [26,59]	**93**	**93**	48	72	69	71	68
B2 CLIP render-and-score	7 [2,22]	31	41	21	62	**100**	70	66
B3 CapFrame [11]	14 [5,31]	45	62	38	52	21	61	53
Ours	**62** [44,77]	72	**93**	**93**	**86**	**100**	**78**	**69**

per scene) and text–image similarity, MLLM judges and a user study as metrics; we use it on the labelled single-object queries and score it against ground-truth objects.

Servo test. Starting from the goal pose perturbed by $0.1\,d^\star$ and 5° (small) or $0.2\,d^\star$ and 10° (large) in three random directions per offset, the same for every desired image so that the methods start from the same relative poses, the photometric servo runs at 320 px width for at most 600 iterations. The desired image is rendered at full resolution and area-downsampled to 320 px, and the current view is rendered at 320 px; at the goal pose itself they differ by a median grey-level MSE of 13–44 per LERF-OVS scene. A run succeeds if it ends within $0.02\,d^\star$ and 1° of the goal pose. The current views are rendered from the same 3DGS model; the supplementary material renders them from an independently trained reconstruction.

4.2 Desired images and servoing

Table 1 compares the methods. Ours gives more good views than CLIP render-and-score and CapFrame (exact McNemar test, $p = 0.004$ and $p = 0.012$), centres the object most often and is always level. The training photo (B1) shows the object more often, since it is the image in which the detector found it. Its good-view rate (21% against 30%) is not significantly lower ($p = 0.18$), and the gap depends on the centring threshold: at 10% of the diagonal, ours has 18 good views against 6; at 20%, 20 against 22. What separates the two is how well the object is framed: on the 29 queries whose object the detector finds, the object centre is 4% of the diagonal from the image centre in our median view, against 16% in the training photo. CLIP render-and-score (B2) rarely shows the whole object (36%): the highest text–image similarity is often reached by a partial or distant view. CapFrame (B3) frames the object at a comparable rate but tilts the camera (median roll 21° over its 61 views; only 10 of the 67 queries get a level view), shows it small, and fails on 6 queries. Most visibility and whole-in-frame failures come from grounding: on the 29 queries whose object the detector finds, our views are whole in 93% of the queries, as many as the training photo, and good

views in 62% against 41% ($p = 0.15$). The visibility test favours the training photo by construction: the ground truth is lifted from labels drawn on training photos, so it holds the surfaces those photos see; for the median instance, one training photo sees 99–100% of its points, while a new viewpoint sees the object from another side. Centring, size and level are also what our selection aims for (it looks at the centroid, sizes the object to 60% of the image and keeps the horizon level); whether they matter is answered by the servo.

The servo reaches our desired images more often than those of CLIP render-and-score and CapFrame for both offsets (Wilcoxon test on per-image success, $p \leq 0.006$), and than the training photo from small offsets (85% against 70%, $p = 0.001$) but not significantly from large ones (76% against 71%, $p = 0.25$). In a logistic regression of the 1572 runs on the criteria (one intercept per method, standard errors scaled for the correlated runs of an image), two properties of a desired image go with success: a centred object (odds ratio 1.75, 95% interval 1.12–2.74) and a short goal distance $d^\star$. They are correlated, and with $\log d^\star$ in the model the effect of centring is no longer significant (1.55, 0.97–2.47); the data do not separate the two. Our views are both centred and close by design (the object fills 60% of the image; median $d^\star$ 0.43 against 0.88–1.42 for the baselines). Since the perturbation and the tolerance are relative to $d^\star$, a distant goal also means a larger absolute motion of the camera among the surrounding objects. Fig. 3 shows examples; videos are on the project page.

How CapFrame fails. CapFrame's intermediate outputs show where its failures in Fig. 3 arise. Part queries are resolved at object level: for "kamaboko" and "bear nose" the subject is the whole ramen bowl or the whole bear, and the camera moves to an oblique, rolled view (54° and 22°) where the part is hidden or cut. Errors propagate along its chain: for "pot" the retrieved view shows a mug in the sink, which is then framed. The pose optimisation constrains neither roll nor the distance to the training views: for "knife" it ends at a rolled pose with coverage 0.01 and a degraded render. These observations come from the 4-bit model; retrieval errors may be fewer at full precision, while roll, drift and object-level subjects follow from the design.

Why the servo fails. Photometric servoing has a narrow convergence basin [12], and most failures come from it rather than from unreachable desired images: only 2 of our 67 desired images are never reached in their 6 runs, success decreases as the initial offset grows, and failed runs stop in a local minimum of the photometric error (median grey-level MSE 1647 against 24 for successful runs). For B1, the desired image is a photograph while the current views are renders; replacing it by the 3DGS render at the photo's pose lowers the residual error at convergence (median MSE 120 to 73) but not the success rate (70% for both offsets), so B1's lower success comes from the viewpoint, not from the photo–render gap. The supplementary material gives the failure statistics per method, the servo on Mip-NeRF 360 and with current views from another reconstruction (success 75% and 67%).

4.3 Grounding versus viewpoint

Many LERF-OVS queries are proper names, brands or dishes ("jake", "dall-e brand", "kamaboko"), outside the detector's vocabulary. Table 2 replaces our zero-shot grounding by LangSplat-V2 [5] masks (its relevancy maps and segmentation rule, from the

Table 2 Stage 1 against viewpoint selection (LERF-OVS, 67 queries, %). Right object: the 3D object centroid lies in the 2–98% bounding box of a ground-truth instance or within 15% of its extent. Good view given the right object, and on the 23 queries both groundings find. LangSplat-V2 returns an empty mask for "pirate hat", counted as a miss.

stage-1 masks	right object	good view (all 67)	good view | right object	good view both right (23)
Grounding DINO + SAM (zero-shot)	43	30	62 (29)	61
LangSplat-V2 [5] (per-scene)	63	34	48 (42)	39
GT label polygon (oracle)	97	69	69 (65)	65
B1 detected training photo	–	21	–	48
B2 CLIP render-and-score	–	9	–	9
B3 CapFrame [11]	–	12	–	13

authors' pretrained per-scene models) and by the labelled polygons, all followed by the same viewpoint selection (Fig. 4). LangSplat-V2 finds the right object more often (63% vs 43%): it distils CLIP features, whose vocabulary is broader, from all views of the scene, and its thresholds were set by its authors on this benchmark; it is not trained on the labels. Given the right object, the detector–SAM masks give more good views (62% vs 48%; 61% vs 39% on the 23 queries both find, 6 queries against 1, $p = 0.13$). This is consistent with SAM masks following the object boundary more closely than thresholded relevancy maps: the object lifted from SAM masks lies more on the ground-truth instance (median 3-D precision 0.89 against 0.68), although the difference is not significant on 23 queries (Wilcoxon $p = 0.17$; supplementary material). With ground-truth masks, 69% of our views are good and 96% show the whole object; 12 of the 21 remaining failures fail only the visibility test (Sec. 5). On the queries both groundings find, CLIP render-and-score and CapFrame stay far below (9% and 13%, $p < 0.005$), and the training photo reaches 48% against 61% for ours ($p = 0.51$).

4.4 Ablation

Table 3 measures the effect on the desired images. The search over candidates matters: taking the PCA normal for every object halves the good views (34% against 62% on the queries whose object is found, $p = 0.008$); the reference-view direction loses 14 points, which is not significant ($p = 0.34$). Removing one score term changes up to 47 of the 67 views (coverage), but the good-view rate moves by at most 7 points, none significant on 67 queries: the terms are partly redundant on this benchmark, where most objects are seen from many training views and fit in free space. They encode constraints that the criteria do not test: coverage keeps the camera where the 3DGS render is reliable (Fig. 2), and the free-space and fit terms keep the camera out of the scene geometry. Changing one parameter by ±50% moves the good-view rate by about 10 points at most, except a larger fill fraction (closer camera, more views cut by the frame), and the continuous prior weight gains 3 queries (not significant; it also drops the planar normal rule); the parameters stay those frozen on the *room* scene. How often each change keeps the selected direction, on LERF-OVS and Mip-NeRF 360, is reported in the supplementary material.

Table 3 Ablation on LERF-OVS (67 queries): number of queries whose selected candidate changes, and good-view rate over all queries and over the 29 queries our grounding finds. A removed score term is replaced by 1 in the score and in the selection rules that use it (normal rule, fit filter, tie-break); ignoring free space places the camera at d_{req}, so that fit is 1 for every candidate; "always the PCA normal" and "always the reference direction" skip the search; the continuous prior weight also drops the planar normal rule. *: different from the frozen method (exact McNemar test, $p < 0.05$).

	variant	views changed	good view (%) all	good view (%) right object
full method	frozen parameters	0	30	62
score term removed	without vis	10	28	59
	without cov	47	27	55
	without π	26	30	62
	without fit	3	30	62
free space	free space ignored	29	31	66
search removed	always the PCA normal	56	18*	34*
	always the reference direction	56	22	48
one parameter ±50%	azimuth step 22.5° (+50%)	35	25	52
	azimuth step 7.5° (−50%)	27	28	59
	fill ϕ 0.85 (+42%, max.)	29	22	45
	fill ϕ 0.3 (−50%)	31	33	72
	margin μ 1.0 (+11%, max.)	9	31	66
	margin μ 0.45 (−50%)	21	25	52
	planarity threshold 0.3 (+50%)	7	30	62
	planarity threshold 0.1 (−50%)	15	31	66
	cov. σ_θ 22.5° (+50%)	16	27	55
	cov. σ_θ 7.5° (−50%)	24	33	69
	cov. $\sigma_{\log d}$ 0.6 (+50%)	22	31	66
	cov. $\sigma_{\log d}$ 0.2 (−50%)	25	28	59
scoring variants	continuous prior weight	16	34	72
	coverage floor 0.25	1	30	62
	both	17	34	72

4.5 A ground robot

The experiments above use a free-flying camera. For a ground robot, only the candidate set changes: the camera stands on reachable floor points around the object, at the robot's height and level, and the score and the selection rule are unchanged. The supplementary material evaluates this variant in 9 synthetic InteriorGS houses [29, 30], with a 3-DOF photometric servo; when the detector finds the object, our view centres it in 69% of the 32 queries against 38% for the survey view in which it was detected (exact McNemar test, $p = 0.04$), but grounding succeeds in only 34% of the 93 queries, and over all queries the difference is not significant. The project page shows the robot navigating to the selected viewpoint and servoing on the desired image.

5 Limitations

The good-view criteria are geometric: they test where the ground-truth object projects and whether the rendered depth occludes it, not whether the render is free of artefacts, so a view far from the training cameras can pass with a degraded image (CapFrame's "knife", Fig. 3). The visibility test ($\geq$50% of the ground-truth points seen) is strict for

volumetric objects, whose ground truth is lifted from labels on several sides, for every method alike. Centring, size and level are also targets of our selection, so the criteria measure how well it achieves them; their value for a robot is supported by the servo results. All experiments run in reconstructed scenes: the current views are renders, not camera images, and a real-robot evaluation is left for future work. Grounding is the main source of failure (Table 2); a language field or a vision-language model could replace the detector. For non-planar objects the selected view is the best supported broadside view, not a semantic front [23]. Transparent, reflective and thin open objects corrupt the rendered depth used to lift the mask (the mask of a bicycle includes the ground seen through its wheels), and the planar decision depends on depth quality: an untextured door is planar ($\rho = 0.014$) with a monocular depth prior during training [31, 32], but not ($\rho = 0.24$) without it. The servo test starts close to the goal (relative offsets in three directions, the same for every desired image), and the desired image is downsampled from a full-resolution render while the current view is rendered at the servo resolution. Reaching a distant desired image needs a servo with a wider basin, e.g. based on matched features, or a sequence of intermediate images; since a 3DGS scene renders any pose, such a visual path could be planned photometrically [33] and ends at our desired image. The CapFrame comparison uses the authors' code with a 4-bit MLLM (the full-precision model does not fit our 24 GB GPU), a templated instruction and, on LERF-OVS, the RGB Gaussians of LangSplat-V2 as scene; its failed queries count as misses.

6 Conclusion

Generating the desired image is the missing step for language-specified visual servoing. A 3DGS scene, zero-shot grounding and a geometric viewpoint selection with free-space and coverage constraints produce desired images that are more often good views than CLIP render-and-score or an MLLM-based text-to-camera method, as often as the training photo in which the object was detected but better centred and always level, and that a photometric servo reaches more often than the images of every baseline from small offsets. Better grounding is the most direct way to improve the pipeline further.

Declarations

- Funding: This research received no external funding.
- Competing interests: The authors declare no competing interests.
- Data availability: All data used in this work are public (LERF-OVS, Mip-NeRF 360, and the InteriorGS scenes distributed with Habitat-GS, used in the supplementary material). Videos and results are on the project page, https://text2desired.github.io/; the benchmark ground truth and results will be released upon acceptance.
- Code availability: The code will be released upon acceptance.
- Author contribution: M.B. wrote the initial code. Y.A. developed the code and its improvements and wrote the manuscript.

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Fig. 3 Desired images for LERF-OVS queries (rows) by each method (columns), on selected queries where our view is a good view, hence better than or similar to the training photo (20 of the 67 queries qualify; results over all queries in Table 1). Green: projected ground-truth object and its centroid; white cross: image centre; ✓/✗: good view or not. The first four rows are queries where the training photo is not a good view (object off-centre or small); in the last two both are. The criteria are geometric and do not test render quality: CapFrame's "knife" view passes them, but its pose is far from every training camera (coverage 0.01) and the render is degraded.

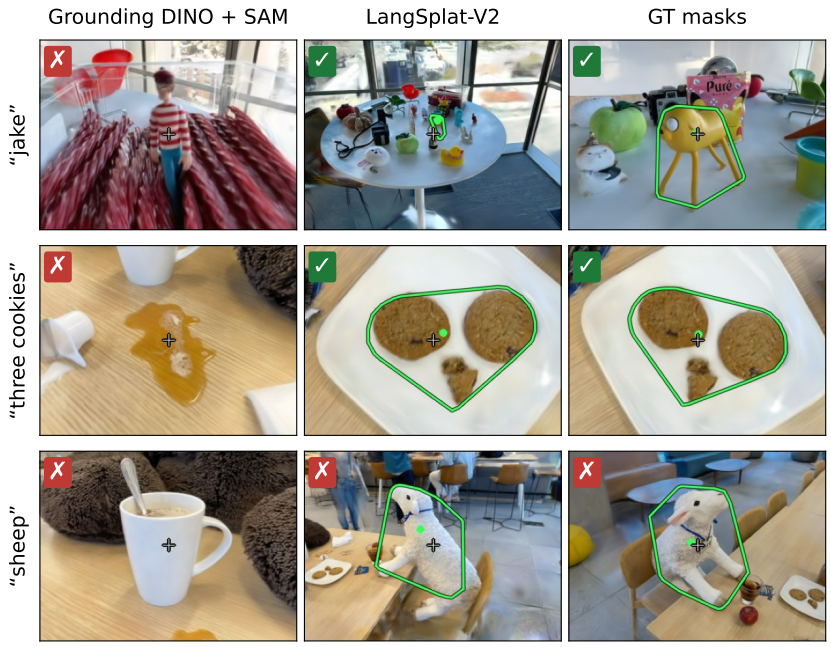


Fig. 4 Grounding failures of the detector on LERF-OVS: our desired images when the stage-1 mask comes from Grounding DINO + SAM, LangSplat-V2 or the ground-truth label, followed by the same viewpoint selection. On "jake" (a figurine) and "three cookies" the detector picks another object (a person in a photo, a stain); with the other masks the view is good. "Sheep" is counted as a failure even with ground-truth masks: the view is whole, centred and well sized, but less than half of the object's surface can be seen from one side (Sec. 5).